Peripheral oxytocin levels are linked to hypothalamic brain volume in adults with Autism Spectrum Disorder

Max-Planck-Institute of Psychiatry

R. Haaf[1,3], ML. Brandi[1], L. Albantakis[1,2,4], L. Henco[1,5], L. Schilbach[1,2,4,5,6].

[1]Independent Max Planck Research Group for Social Neuroscience, Max Planck Institute of Psychiatry, Munich, Germany, [2]Outpatient and Day Clinic for Disorders of Social Interaction, Max Planck Institute of Psychiatry, Munich, Germany, [3]Graduate School Technical University of Munich, Munich, Germany, [4]International Max Planck Research School for Translational Psychiatry, Munich, Germany,[5]Graduate School of Systemic Neurosciences, Munich, Germany, [6]Ludwig-Maximilians-Universität München, Munich, Germany

MAX-PLANCK-GESELLSCHAFT

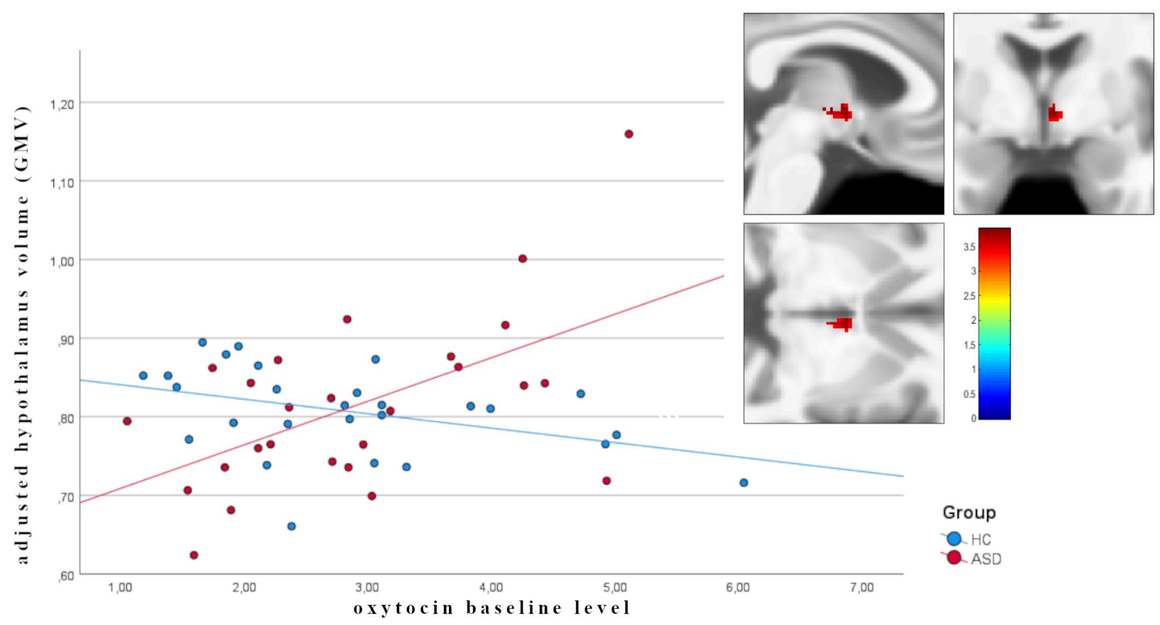
1 Introduction:

Oxytocin (OXT) is known to modulate social behavior and cognition and, thus, expected to be a potential therapeutic agent for individuals with autism spectrum disorder (ASD). However, the underlying neurobiology is far from clear. For instance, relatively little is known about the association between OXT levels and brain structural differences in ASD patients. Here, the hypothalamus (HTH) is known as the key regulatory site of the OXT system. Morphometric studies in ASD patients have, indeed, reported a reduction in HTH volume in children and adolescents (reviewed in Caria et al.,2020). However, the relationship between HTH volume and peripheral OXT levels has not been verified by actual OXT measurements, nor has this link been studied in adults with ASD. We, therefore, assessed HTH morphometry using voxel-based morphometry (VBM) and peripheral OXT levels in a sample of high-functioning adults with ASD.

2 Methods:

Structural T1-weighted MRI images were obtained on a 3T GE scanner in 29 ASD patients (36.03±11.0 years m:17,f:12) and 27 matched controls (30.96±11.2 years m:9,f:18). Images were pre-processed and analysed using the SPM12-based toolbox for VBM analyses Cat12 (http://www.neuro.uni-jena.de/cat/). Smoothed and modulated grey matter images were entered in a region of interest (ROI) analysis using a HTH mask (Pauli et al.,2018). Results of the voxel-wise group comparison and correlation analyses were assessed at a Family-Wise-Error (FWE) corrected level of $p<0.05$ (with an initial peak-threshold of $p<0.001$ and subsequent cluster-level inference). For correlation analyses with peripheral OXT levels, plasma samples at rest were used (protocol described in Albantakis et al.,2021). Autistic traits were assessed by using the Autism-Spectrum Quotient (AQ) questionnaire (Baron-Cohen et al.,2001). All analyses included total intracranial volume (TIV), age and sex as nuisance covariates.

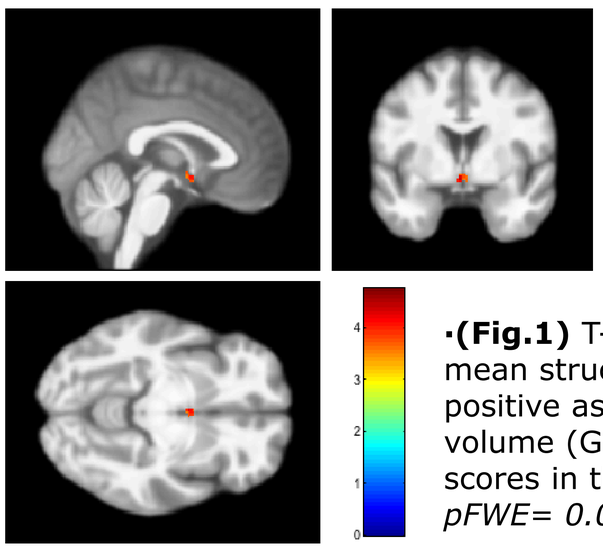
3



·**(Fig.2)** T-score projection and scatterplot for the group comparison of associations between GMV of the HTH and OXT levels with the contrast [GM [ASD] × OXT > GM [HC] × OXT] (k=46, *pFWE=0.017*)

4 Results:

Voxel-wise group comparison of grey matter volume (GMV) in the HTH did not show significant group differences. Separate correlation analysis in both groups, however, showed that GMV in the HTH was positively associated with AQ scores in the ASD group (*FWE-corr. p=0.014*), while no correlation was found in the control group (Fig.1). Group comparison of the link between GMV and OXT levels showed a significant positive association in the ASD group as compared to a negative association in the HC group (*FWE-corr. p=0.017*) (Fig.2).



·**(Fig.1)** T-score projection on mean structural image for the positive association of grey matter volume (GMV) of the HTH and AQ scores in the ASD group (k=46, *pFWE= 0.014).*

5 Conclusion:

Based on our results, HTH brain volume does not differ between adults with and without ASD. However, we did observe a link between interindividual differences in autistic traits and HTH brain volume, but only in the patient group. In addition, in patients as compared to controls we observed positive correlation between peripheral OXT levels and HTH brain volume. These findings are consistent with previous research indicating that specific volumetric differences in ASD may only be apparent at certain developmental stages at the group level (Courchesne et al.,2011), but shed new light onto the role of interindividual differences in this regard. Furthermore, we found that peripheral OXT levels in patients are closely linked to volumetric HTH differences. Although limited by its relatively low sample size, this study does suggest a potentially important role of the hypothalamus in the pathophysiology of ASD and its relationship to the OXT system.

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